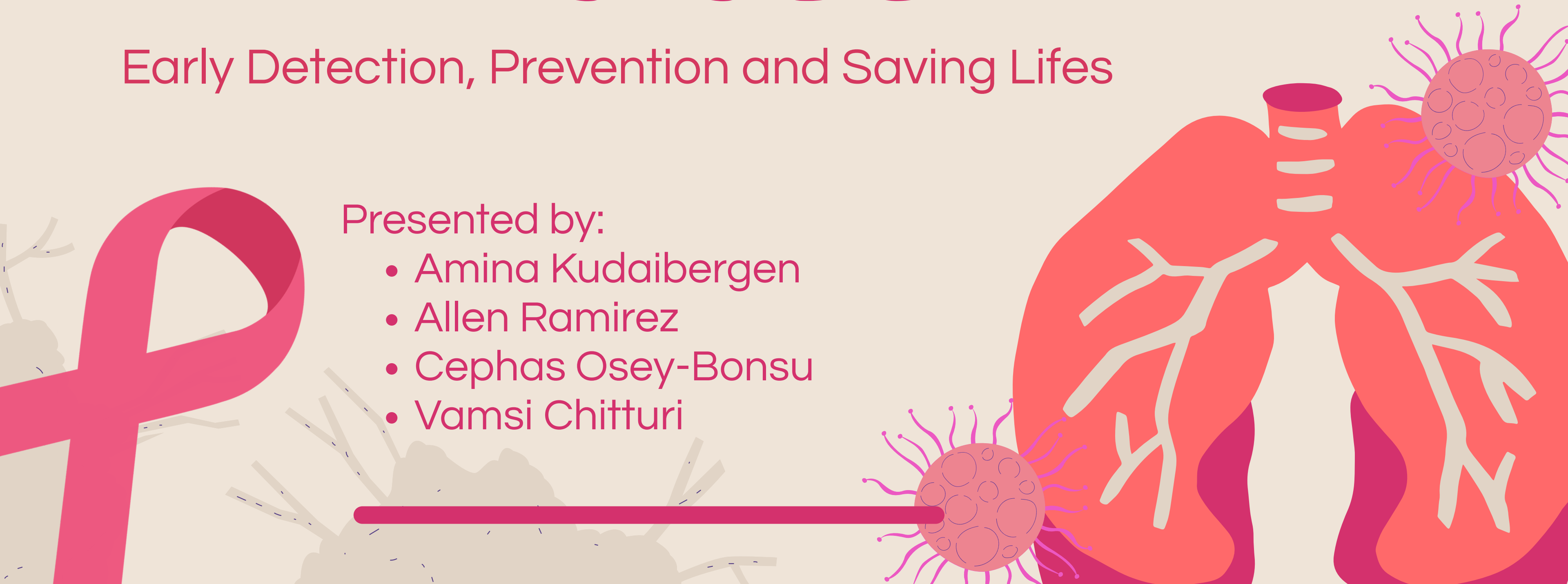


AI PATHOLOGIST

Early Detection, Prevention and Saving Lives

Presented by:

- Amina Kudaibergen
- Allen Ramirez
- Cephas Osey-Bonsu
- Vamsi Chitturi





Detected early, breast cancer has a 5-year survival rate of over 99%; once it spreads to distant organs, that rate falls to 31%. (American Cancer Society, SEER Data, 2026)



Breast cancer survival exceeds 90% in high-income countries but drops to just 40% in South Africa. (World Health Organization, 2025)



Breast cancer incidence among women under 50 is rising 1.4% per year, nearly double the rate in older women. (American Cancer Society, 2024)





Breast cancer remains a leading cause of death among women, worsened by delayed diagnosis and limited healthcare access in low and middle income countries.



Our project is an explainable AI prototype that classifies breast histology images as Benign, Carcinoma In Situ, Carcinoma Invasive, or Normal.



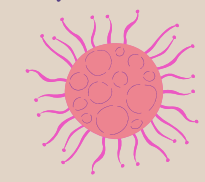
It uses an ensemble of four CNN models, ResNet50, VGG16, DenseNet121, and EfficientNet-B0, with Grad-CAM heatmaps highlighting the regions behind each prediction.



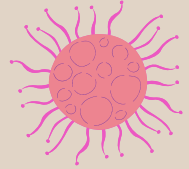
The ensemble achieved 86.75% accuracy



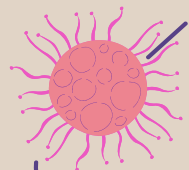
RESEARCH QUESTION



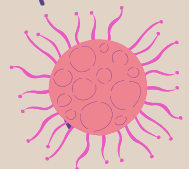
Can we accurately classify tissue as normal, benign, or a type of malignant, while highlighting regions that influenced these decisions with a 90% certainty?



Normal: No tumors and lesions with a healthy architecture

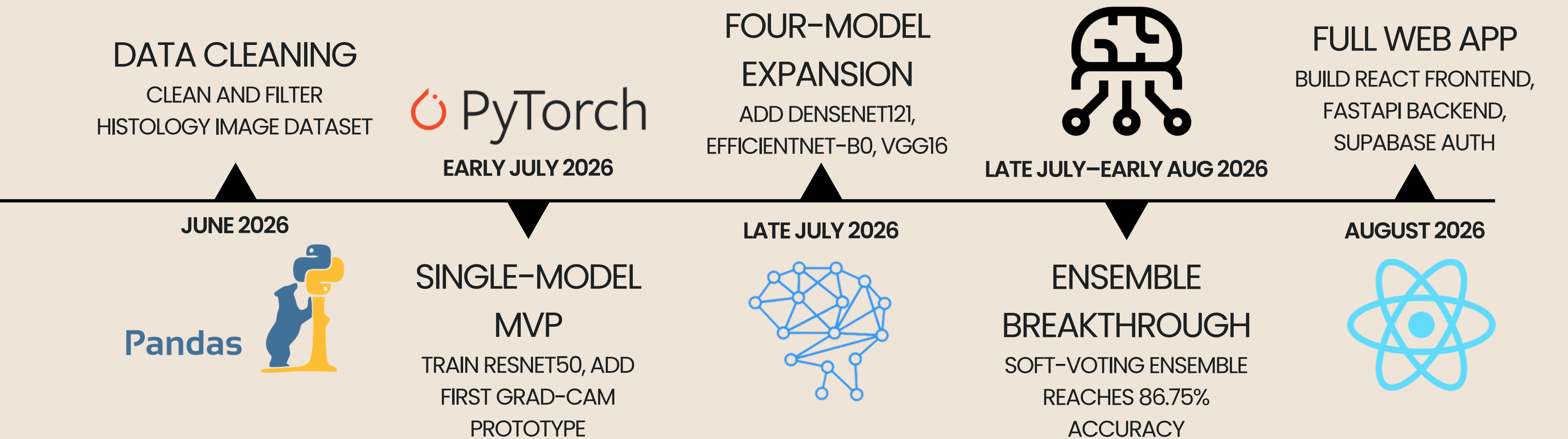


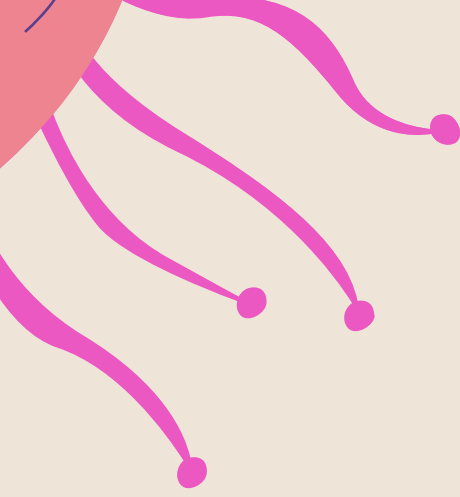
Benign: Non cancerous tumors with minor lesions



Malignant: Cancerous tumors with major lesions

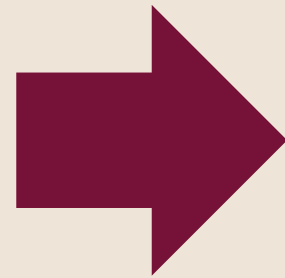
HOW OUR PROJECT EVOLVED IN THIS PROGRAM?





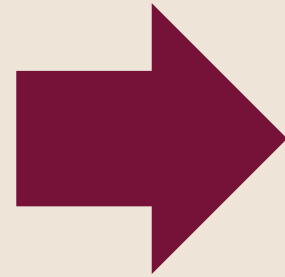
HOW OUR PROJECT FACED CHALLENGES

Single-Model Limitations



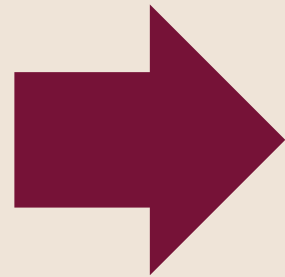
Using only ResNet50 capped accuracy at 75%, so we combined four models into a soft-voting ensemble to reach 86.75%

Black-Box Predictions



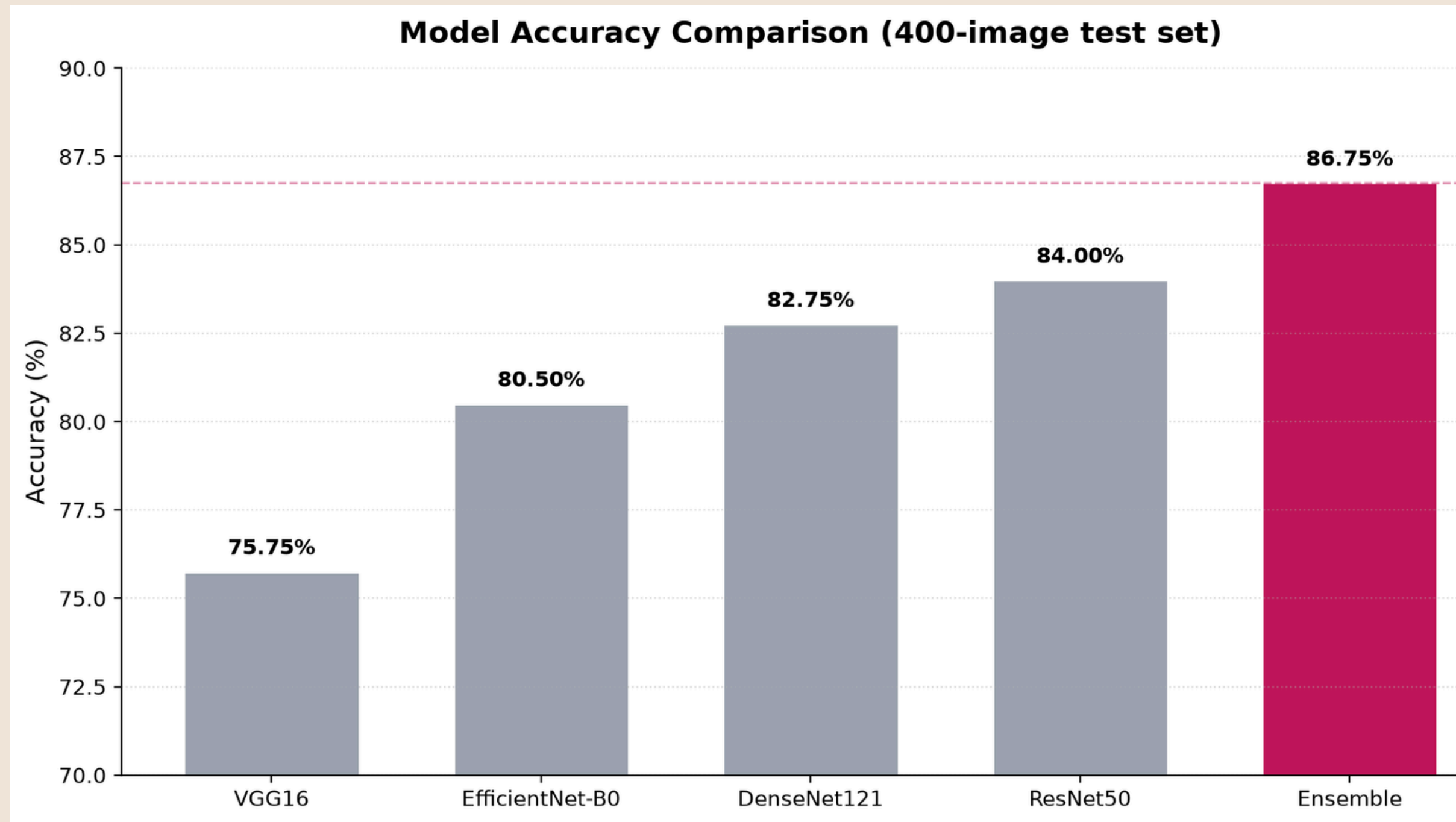
Raw class labels are hard to trust in medical AI, so we added Grad-CAM heatmaps to show which regions influenced each prediction.

ML Deployment Constraints



Large model weights couldn't run on standard serverless hosting, so we separated the frontend, auth, and ML backend into dedicated services.

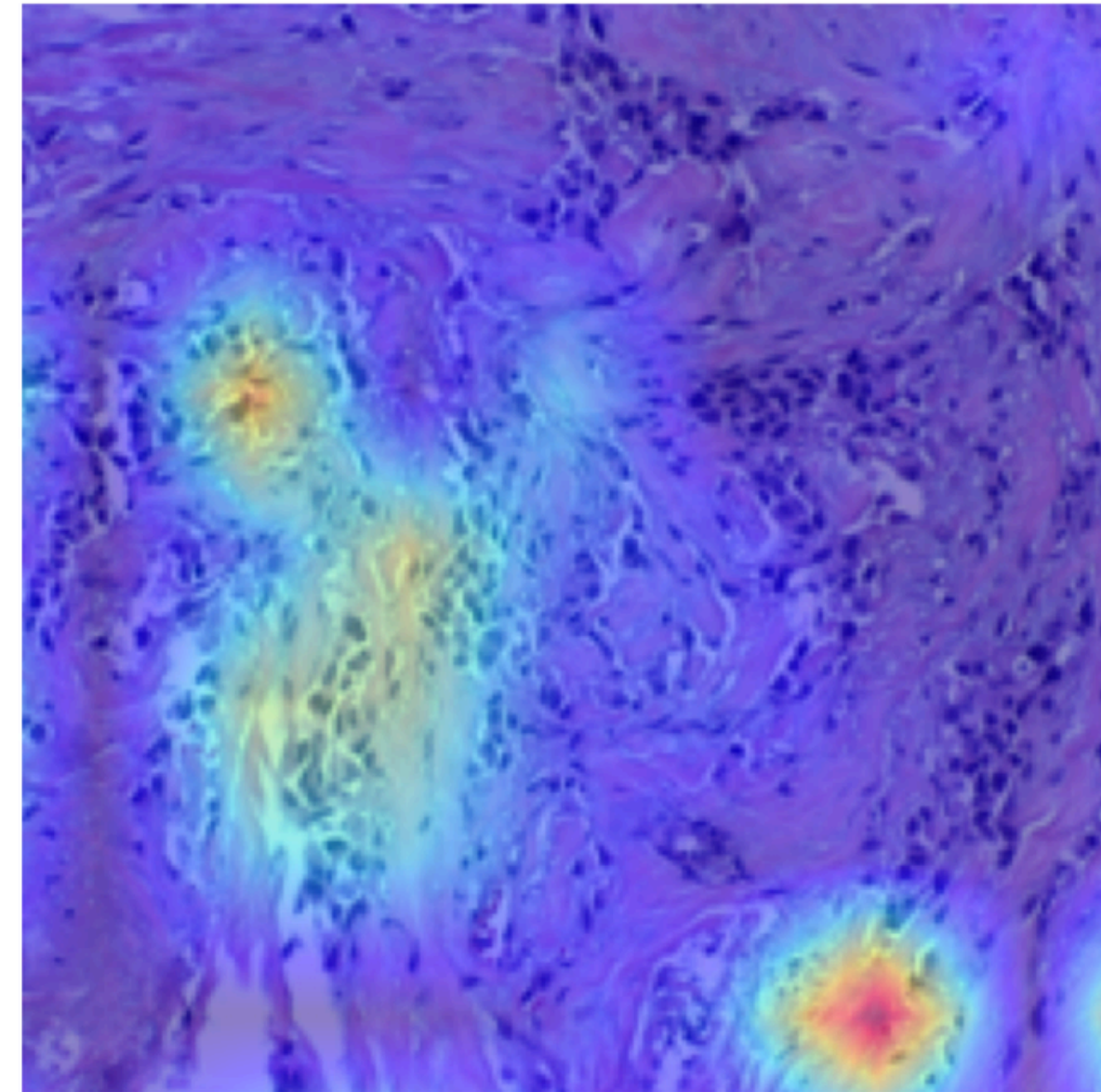
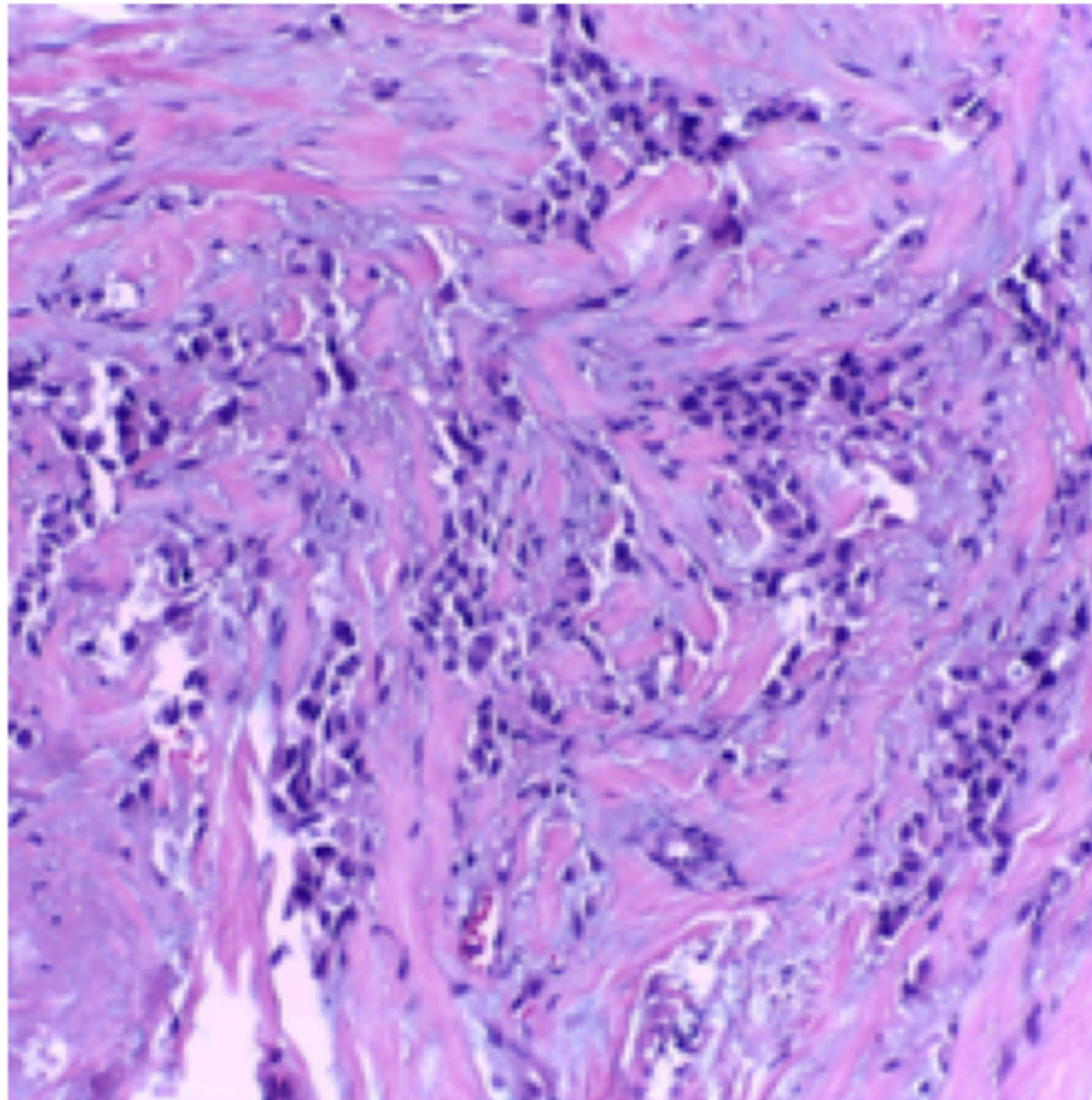
DATA VISUALIZATION #1



Combining four models raised accuracy from 75% to 86.75%, directly solving our single-model limitation.

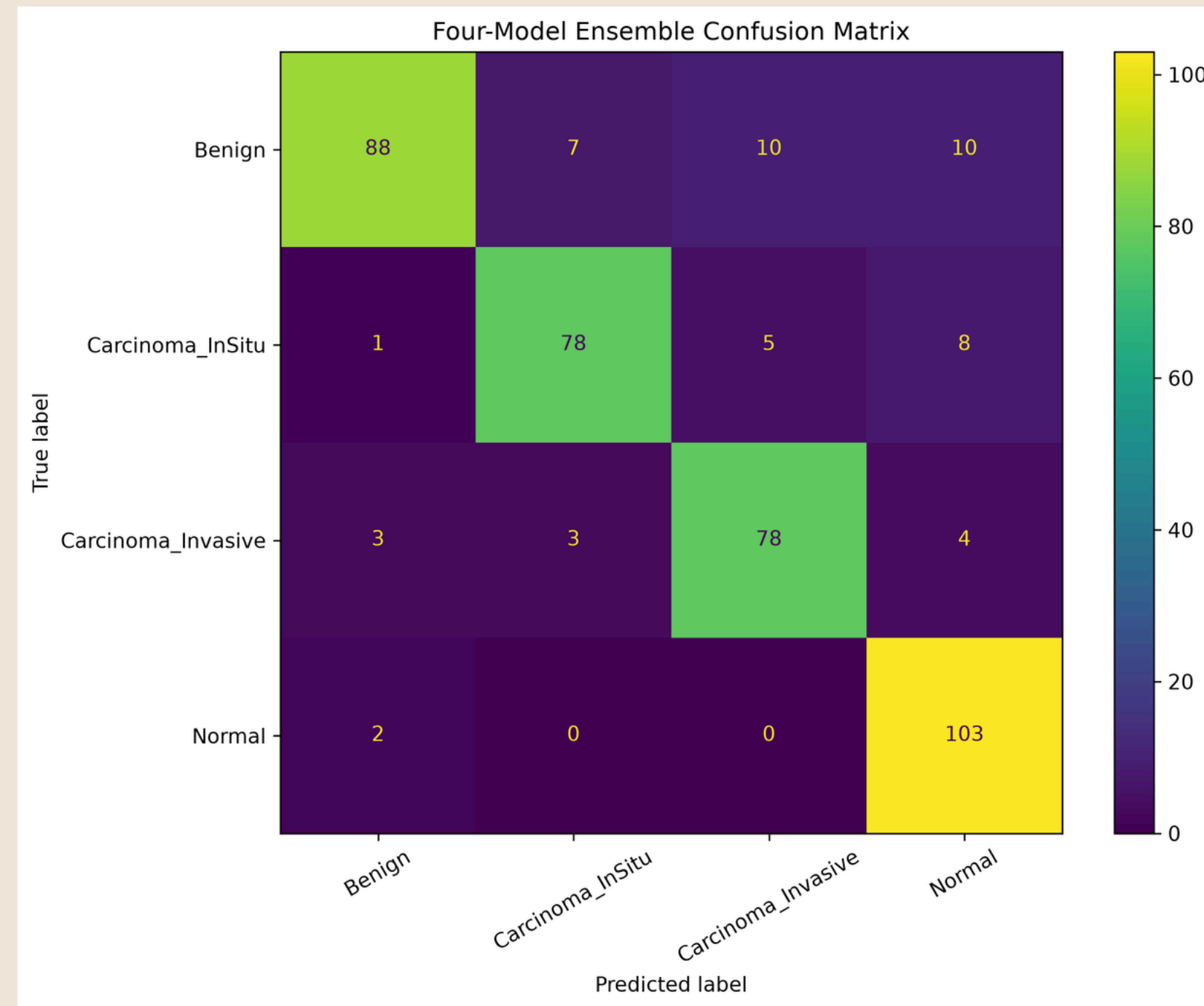
DATA VISUALIZATION #2

Explainability: What the Model Focused On
Original Histology Image **Grad-CAM Overlay**



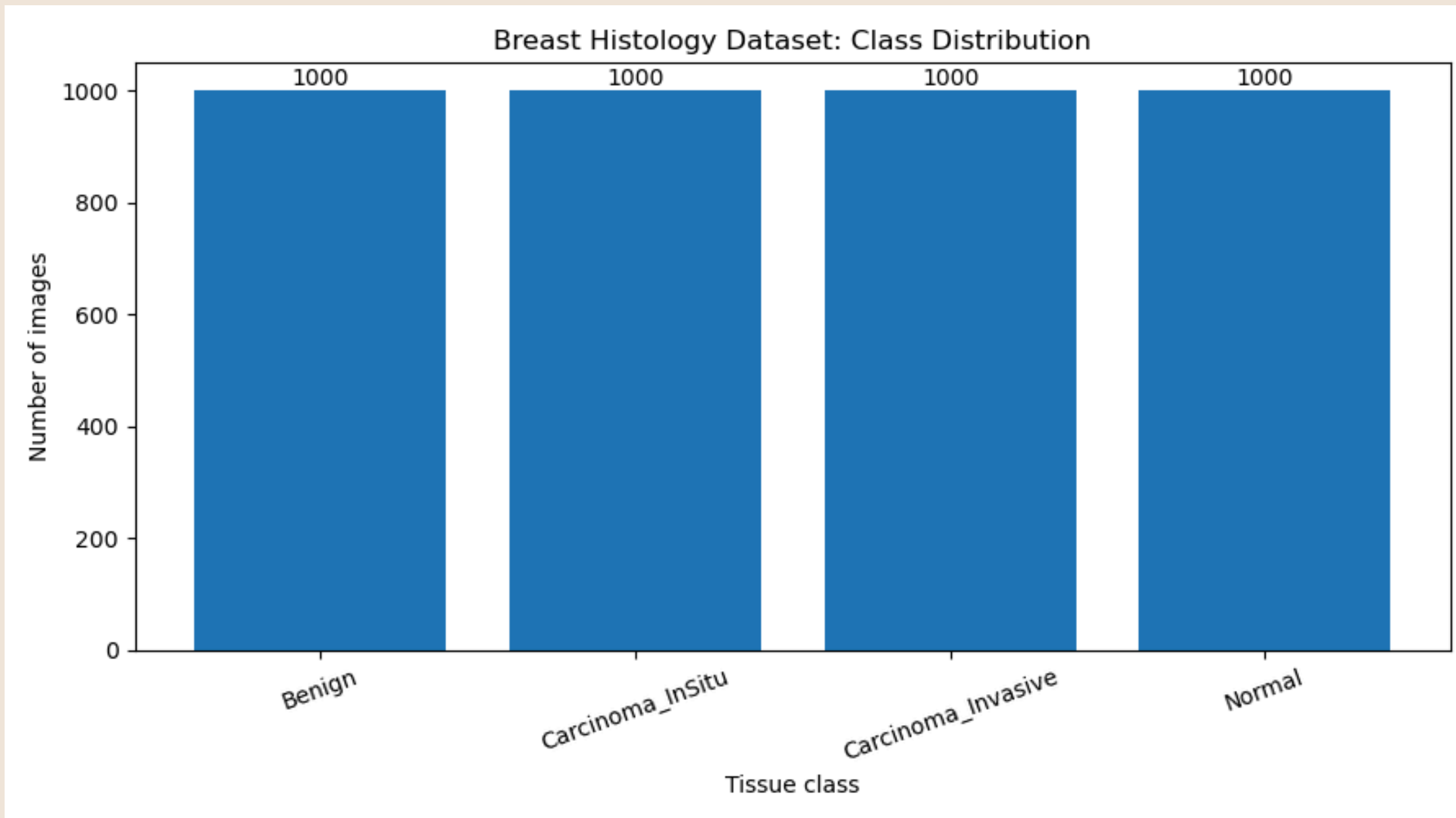
Grad-CAM heatmaps reveal exactly which regions drove each prediction, turning a black box into an explainable result.

DATA VISUALIZATION #3



The confusion matrix shows where our deployed ensemble still struggles, guiding where expert review matters most.

UNDERSTANDING OUR DATA

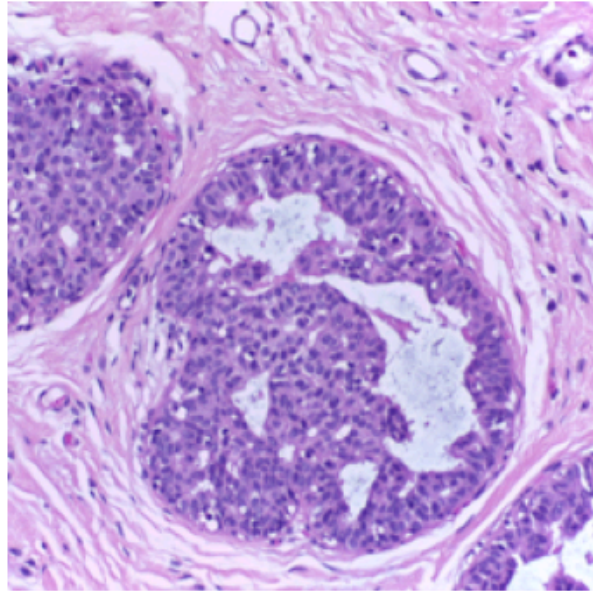


Our dataset is evenly balanced across all four classes, reducing bias toward any single diagnosis. 4,000 training images total.

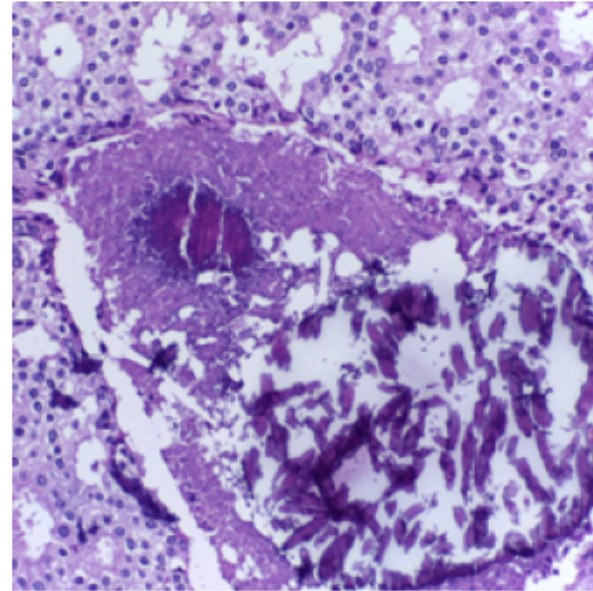
UNDERSTANDING OUR DATA

Sample Breast Histology Images

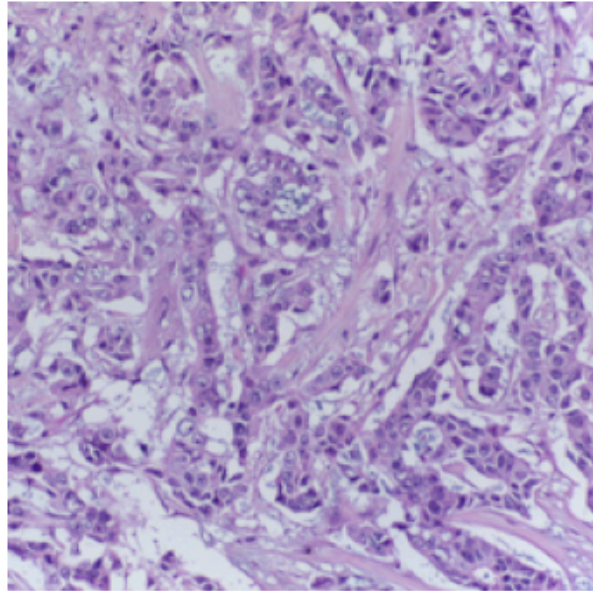
Benign



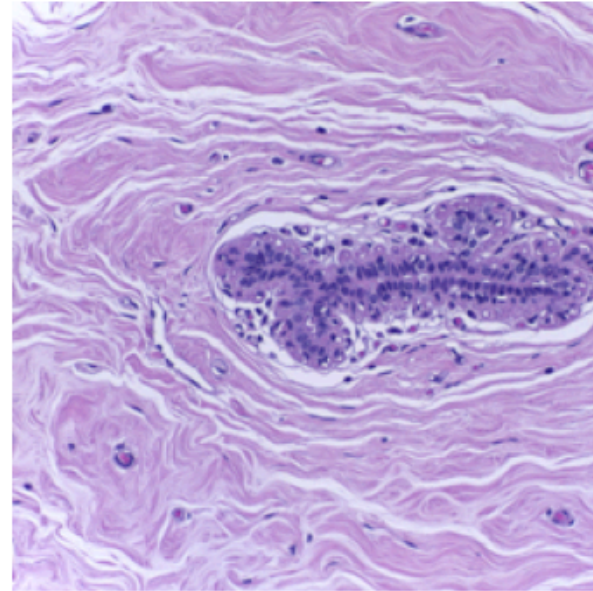
Carcinoma_InSitu



Carcinoma_Invasive

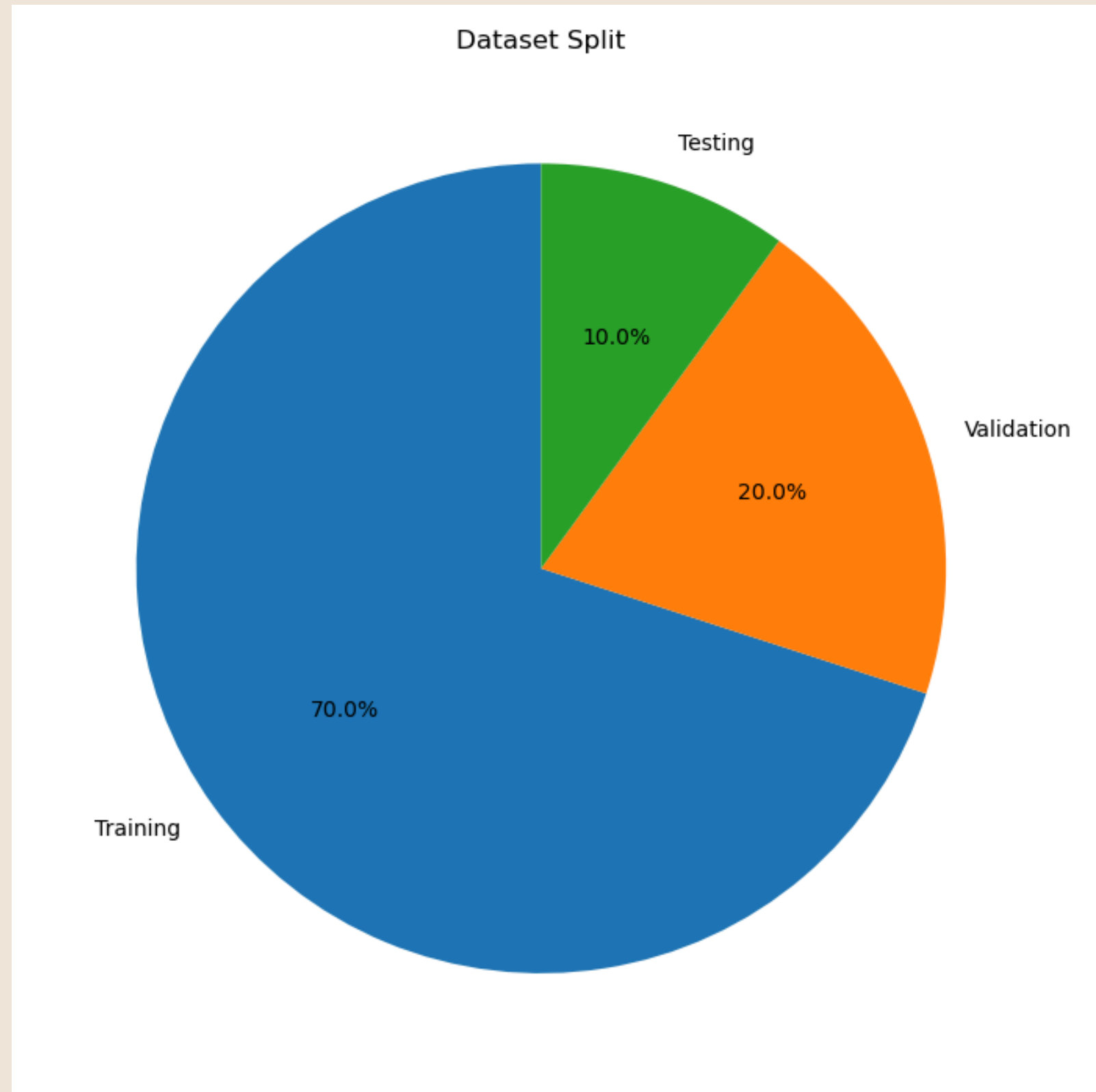


Normal



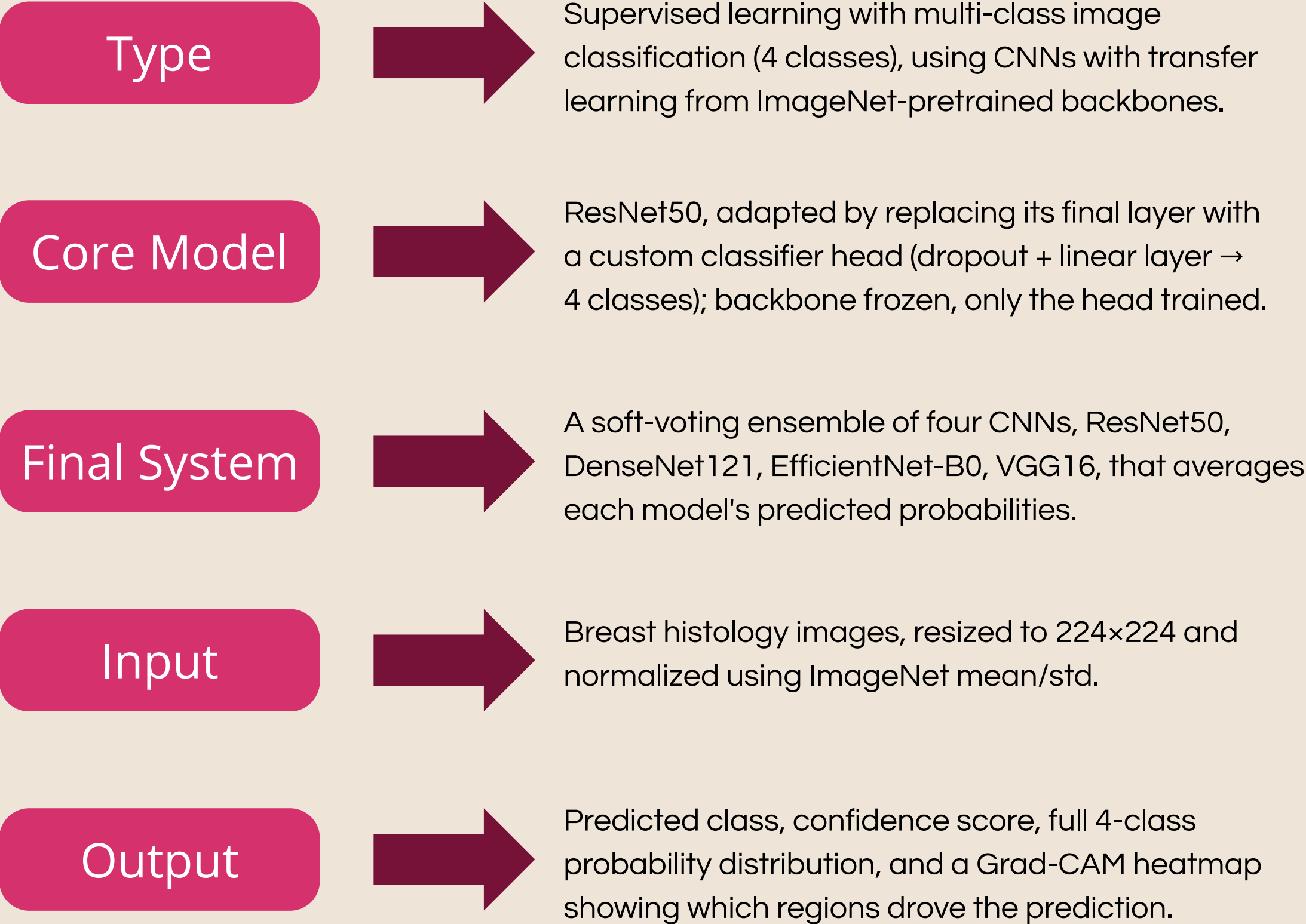
Each class has a distinct visual signature, which is what our models learn to detect.

UNDERSTANDING OUR DATA



We reserved 30% of the data for validation and testing to ensure our reported accuracy reflects real performance, not memorization. Validation is necessary as it provides an unbiased evaluation of a model fit on the training dataset while tuning the model's hyperparameters

THE ALGORITHM



Optimizer	AdamW
Learning Rate	1e-4
Batch Size	16
Epochs	8
Loss	CrossEntropyLoss

```
RESNET50 HEAD REPLACEMENT

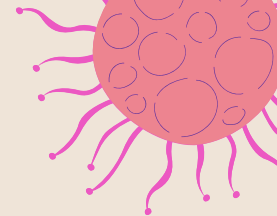
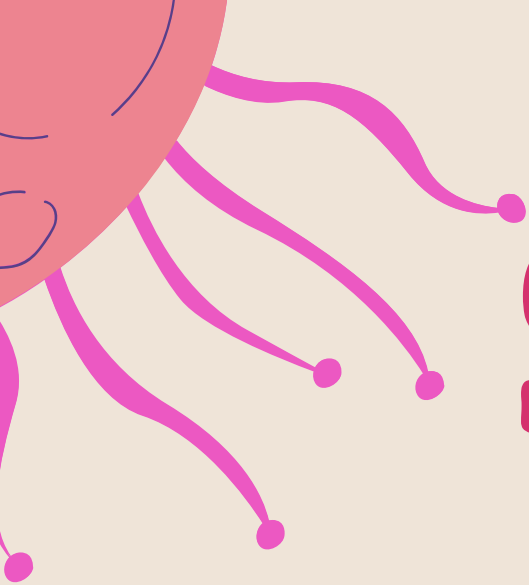
model =
  resnet50(weights=ResNet50_Weights.DEFAULT)
model.fc = nn.Sequential(
  nn.Dropout(0.3),
  nn.Linear(model.fc.in_features, 4)
)
```

HOW DATA SHAPED THE ALGORITHM

Data Trait	Why It Mattered	Our Response
Small Balanced Datasets (4,000 Images & 4 Classes)	Training a large CNN from scratch on this size of dataset risks overfitting.	Used transfer learning with a frozen, ImageNet-pretrained backbone instead of training from scratch.
Subtle visual differences between classes	Benign, in-situ, and invasive tissue can look visually similar in places; one architecture may miss features another catches.	Combined four CNNs (ResNet50, DenseNet121, EfficientNet-B0, VGG16) into a soft-voting ensemble.
Uneven recall despite balanced training data (Benign: 76.5%)	A single confident label can hide borderline or ambiguous cases, risky in a medical context.	Return the full 4-class probability distribution plus a Grad-CAM heatmap, not just the top label.

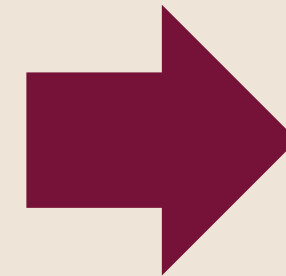
PROS & CONS OF OUR APPROACH

Component	Pros	Cons
ResNet50	Best single-model accuracy (84%); mature, well-suited to transfer learning.	Alone insufficient for final use; still a black box without Grad-CAM.
Transfer Learning	Faster training; strong performance despite limited medical data.	ImageNet features are natural-image based; frozen backbone may miss medical detail.
4-Model Ensemble	Raised accuracy from 84% to 86.75%; more robust than any single model.	Slower inference, higher compute/memory, and harder to deploy.
Grad-CAM	Builds trust by showing which regions influenced each prediction.	Coarse explanation map, not a diagnosis; can mislead if over-interpreted.



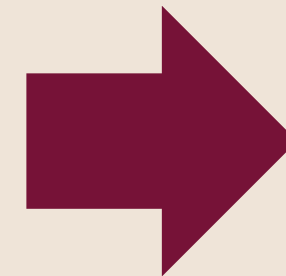
SOURCES OF BIAS & SOLUTIONS

Underrepresentation of
Cancer Classes



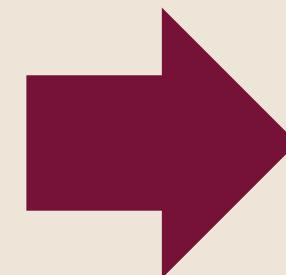
Data has more images of one class than another class (Walsh & Tardy)

Lack of Age Stratified Data



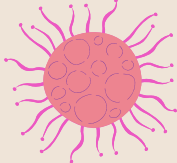
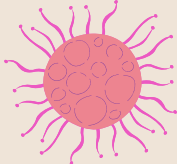
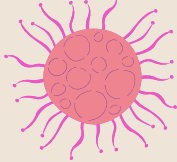
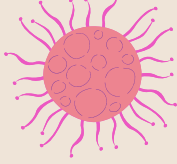
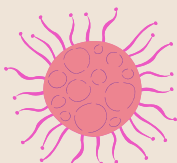
Data does not account for how the disease looks across different age groups (Kramp et al.)

Lack of Staining Variability

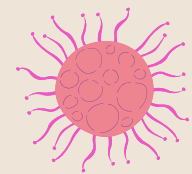


Histology images come from specific laboratories with a different spectrum of colors (Ochi et al.)

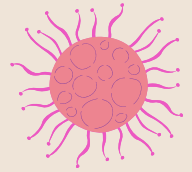
ETHICAL IMPLICATIONS

-  Data can be un-ethically sourced without the consent of patients
-  New user data can be un-ethcially used to train future models
-  AI/ML model can provide in-accurate diagnosis
-  Patient's privacy risk on medical history and personal identification
-  If the data used to train models is not representative of the population it may misguide patient's treatment.

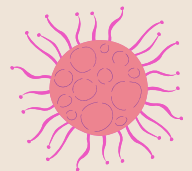
ESSENTIAL QUESTIONS



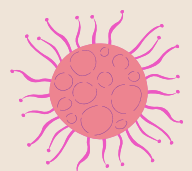
HOW CAN AI BE USED RESPONSIBLY IN HEALTHCARE?



AI is only responsible in healthcare when it stays transparent and reviewable, which is why we built Grad-CAM so predictions can be checked, not blindly trusted.

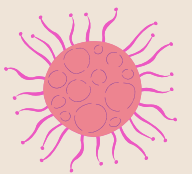
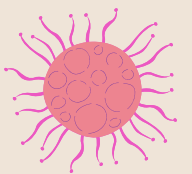
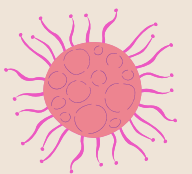
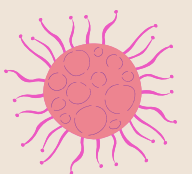


It also requires controlled access, since open, unauthenticated tools risk misuse; we addressed this with Supabase authentication requiring sign-up and sign-in.



Ultimately, responsible AI means pairing technical capability with human oversight, not deploying a black-box tool alone.

NEXT STEPS

-  Next, we will improve Benign recall through class-weighted loss, oversampling, and expanded fine-tuning BY AUGUST 15TH 2026
-  Next, we will finalize dataset documentation and validate on external, unseen data sources BY AUGUST 20TH
-  Next, we will add model-disagreement flags and low-confidence warnings to strengthen clinical trust BY SEPTEMBER 2026
-  Next, we will deploy the ML backend on always-on cloud infrastructure for reliable production use. BY END OF SEPTEMBER 2026

PEER REVIEW

“It is has a nice UI and is easy to use and understand”

-Norah

“Using an ensemble of 4 powerful models is exactly what the problem needs”

-Amirkhan

“I also used a similar idea for my hackathon project for all cancer types, but we did not think of adding heatmaps”

-Joseph

“I would add a login and registration page for added security”

-Amaya

“This project is so inspirational; I have never built an ML full-stack application before, and it makes me want to make one!”

- Abdelrahman

CITATIONS

American Cancer Society — Breast Cancer Survival Rates:

<https://www.cancer.org/cancer/types/breast-cancer/understanding-a-breast-cancer-diagnosis/breast-cancer-survival-rates.html>

World Health Organization — Breast Cancer Inequities / Global Breast Cancer Initiative:

<https://www.who.int/initiatives/global-breast-cancer-initiative/breast-cancer-inequities>

HEROZA / Lince-Deroche et al. — Costs of Breast Cancer Diagnostics in Johannesburg, South Africa:

<https://www.heroza.org/wp-content/uploads/2019/09/Lince-Deroche-et-al.-Costs-diagnostics-in-Joburg-RSA-working-paper-2019.....pdf>

Cancer Research UK. (2026, February 13). *Why is early cancer diagnosis important?* <https://www.cancerresearchuk.org/about-cancer/spot-cancer-early/why-is-early-diagnosis-important>

Deep learning empowered breast cancer diagnosis:

Advancements in detection and classification. PLoS One. 2024 Jul 11;19(7):e0304757. doi: 10.1371/journal.pone.0304757. PMID: 38990817; PMCID: PMC11239011.

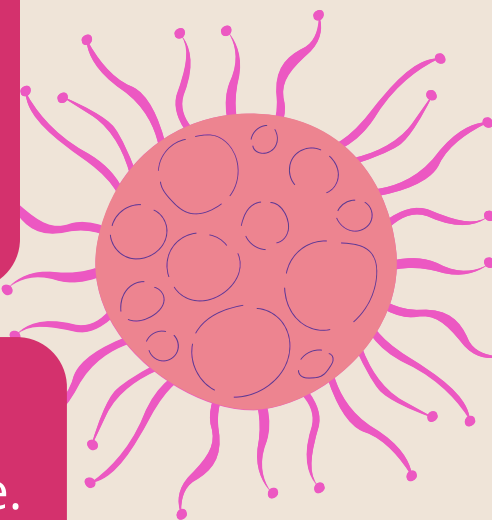
National Breast Cancer Foundation. (n.d.). Early detection of breast cancer. National Breast Cancer Foundation.

<https://www.nationalbreastcancer.org/early-detection-of-breast-cancer/>

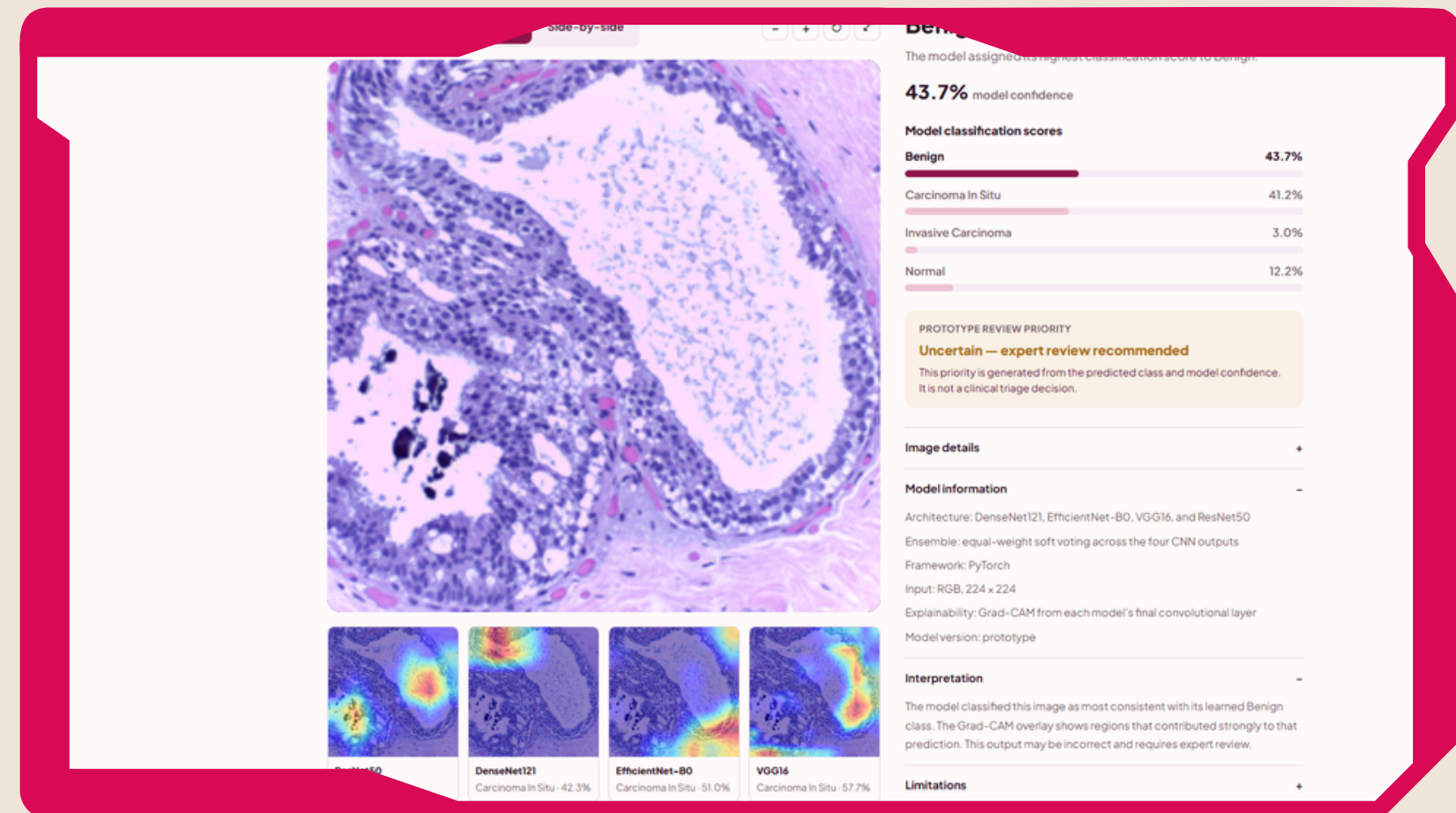
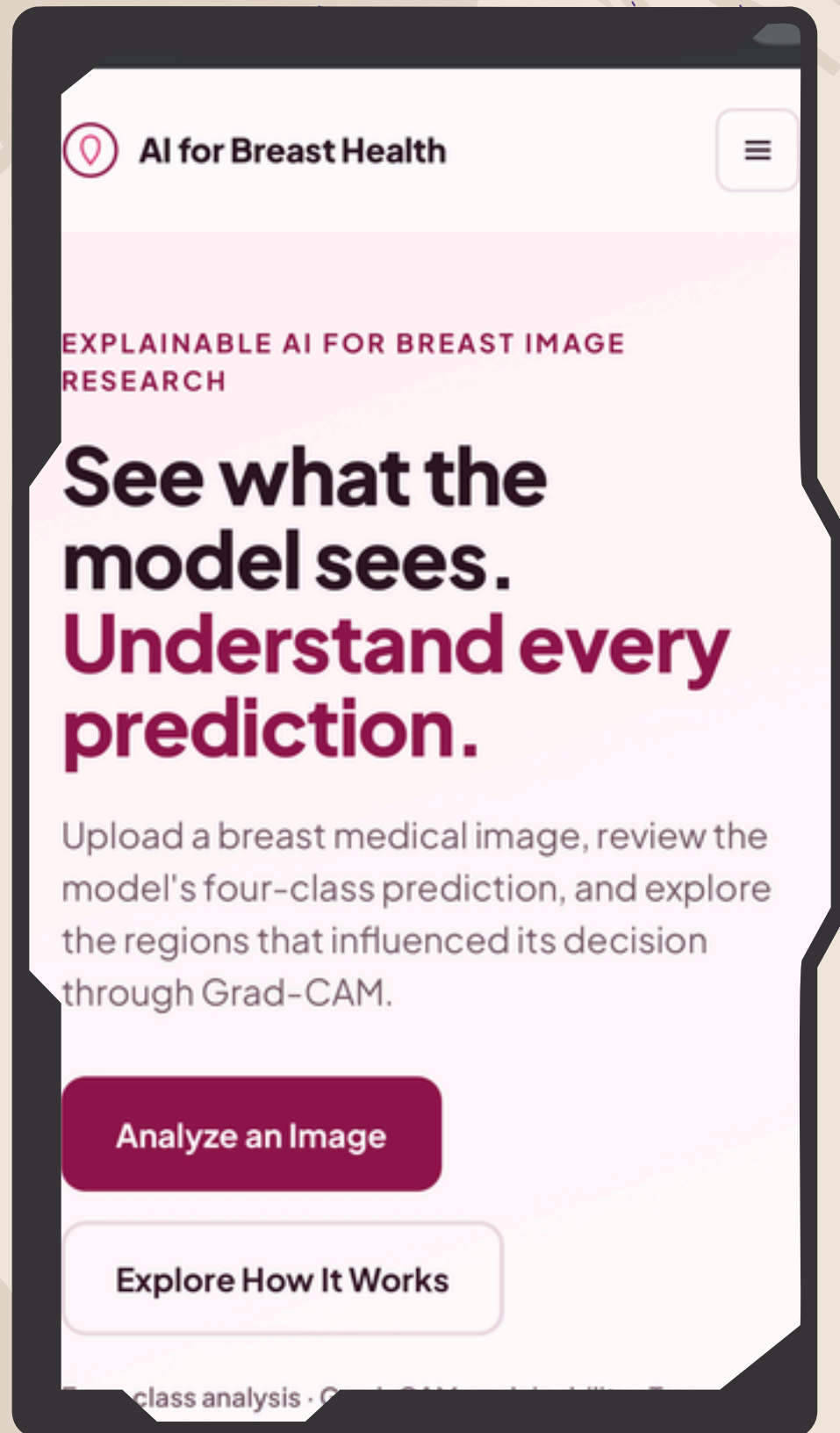
Jiang, B., Bao, L., He, S., Chen, X., Jin, Z., & Ye, Y. (2024). Deep learning applications in breast cancer histopathological imaging: diagnosis, treatment, and prognosis. Breast Cancer Research, 26, 137. <https://doi.org/10.1186/s13058-024-01895-6>

Ekpo, E. U., Alakhras, M., & Brennan, P. (2018). Errors in mammography cannot be solved through technology alone. Asian Pacific Journal of Cancer Prevention, 19(2), 291–301.

<https://doi.org/10.22034/APJCP.2018.19.2.291>



DEMO TIME



THANK YOU

Presented by AI Pathologist

